A Fully-Parallel Digital MRAM Computing-in-Memory Macro Featuring a High-Efficient Dynamic Adder Tree and Bit-Splitting MAC

Zhongzhen Tong[1,2†], Jiye Yao[1†], Shaohui Ma[1], Yulong Qiu[1], Zhaohao Wang[1,2], Amara Amara[1], Xiaoyang Lin[1,2*]
[1]*State Key Laboratory of Spintronics, Hangzhou International Innovation Institute, Beihang University, Hangzhou 311115, China.* [2]*School of Integrated Circuit Science and Engineering, Beihang University, Beijing 100191, China.* **Corresponding Author Email: xylin@buaa.edu.cn.* [†]*These authors contributed equally to this work.*

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***Abstract*—Digital computing in-memory (DCIM) macros, known for their high robustness, have recently attracted growing attention. However, they still face challenges of large logic overhead and low efficiency sign-bit processing in fully-parallel CIM. In this work, we propose a fully-parallel digital MRAM-CIM architecture featuring a dynamic adder tree and a bit-splitting multiply-and-accumulate (MAC) scheme. Based on a 28 nm CMOS process and voltage-gated spin-orbit torque magnetic tunnel junctions (VGSOT-MTJs) devices, a 128-kb MRAM-CIM macro was simulated to validate the proposed design. The dynamic full-adder and 15-to-4 full-precision compression-based adder tree achieves 48.7% lower energy for MAC operations, while the bit-splitting approach eliminates sign-extension operations, improving computing efficiency. Furthermore, the shared voltage-divider reference cell enables compact 2T1M storage, achieving 73.96% and 64% area reduction compared with complementary cell designs. The proposed MRAM-CIM achieves energy efficiency of 96.9 TOPS/W and throughput of 2.8 TOPS at IN-8b/W-8b/OUT-22b with 0.9 V VDD. Its figure-of-merit is 3.03–5.27× higher than existing CIM macros.**

***Index Terms*—Computing in-memory (CIM), Dynamic Adder Tree, Bit-Splitting MAC, MRAM, VGSOT-MTJ.**

I. Introduction

With the rapid advancement of artificial intelligence, computational demands have risen dramatically. Traditional architectures constrained by the von Neumann bottleneck can no longer meet these requirements, driving the rapid development of computing in-memory (CIM) [1], [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15] architectures. Among them, Digital CIM (DCIM) [1], [10], [11], [15], [16], [17], [18], [19], [20] has garnered increasing attention in recent years for its high robustness and process scalability. However, as illustrated in **Fig. 1**, DCIM still faces several critical challenges.

1) Large overhead of adder tree: In DCIM architectures, the adder tree dominates both area and energy consumption [1], [8], [10], [11]. Conventional CMOS-based adder trees typically employ 1-bit adders in the first stage and 2-bit adders in the second, requiring up to 127 half adders (HAs) and 120 full adders (FAs) for a 64-accumulation length. This large overhead arises from non-saturated logic between stages. For instance, when all first-stage inputs are logic '1', the output "10" propagates to the next 2-bit adder stage, where two "10" inputs yield "100". While a 2-bit adder can represent up to "111", only "100" is utilized, causing substantial logic redundancy. Moreover, CMOS logic further increases the area overhead [1].

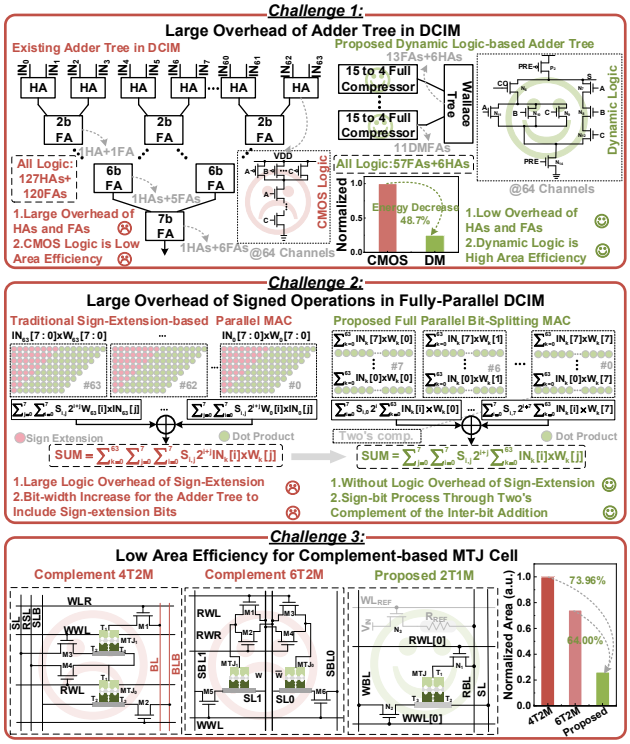


Fig. 1. The challenges in existing CIM architectures.

2) Large overhead of signed operations: DCIM can be implemented in either bit-serial [8], [15], [17] or fully-parallel [5], [6] modes. Bit-serial computation offers high area efficiency, as signed operations can be handled via two's complement on partial product sums [8]. However, it requires multiple cycles, such as eight for an 8-bit input. Fully parallel computation completes the entire CIM operation in a single cycle but demands additional sign-extension logic for INT-8b multiplication [5]. This not only increases circuit complexity but also increases the bit width of the subsequent adder tree, resulting in additional hardware overhead.

3) Low area efficiency for complement-based MTJ cell [1], [17], [18], [21], [22], [23]*:* Because of the limited tunneling magnetoresistance (TMR) of magnetic tunnel junctions (MTJs), achieving large read margins remains a challenge. Complementary storage schemes, such as the 4T2M voltage-divider cell [1], [17] and 6T2M discharge cell [18], improve read reliability but store only one bit using two MTJs, resulting in reduced density.

To overcome these limitations, this work proposes an MRAM-CIM macro based on a high-efficiency dynamic adder

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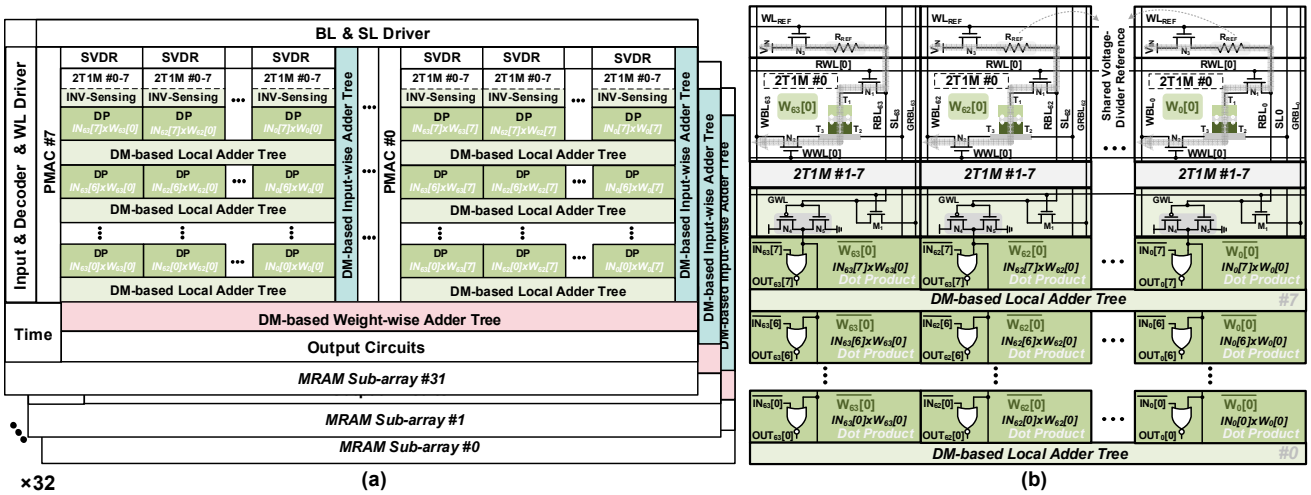


Fig. 2. (a) The overall architecture of the proposed MRAM-CIM. (b) The schematic of mini-MRAM array.

tree and bit-splitting multiply-and-accumulate (MAC) architecture. The key contributions are summarized as follows:

1. The proposed dynamic local adder tree (DMLAT) employs a 15-to-4 full-precision compression (FC) in the first stage and a custom Wallace tree in the second, achieving full-bit logic utilization for both inputs and outputs. Consequently, a 64-accumulation length adder tree requires only 57 FAs and 6 HAs. The use of dynamic FAs (DM-FA) in FC further enhances energy efficiency, yielding a 48.7% reduction in energy consumption.
2. The bit-splitting MAC technique requires two's complement only for inter-bit addition, effectively eliminating the need for sign-extension logic and reducing computational overhead.
3. A shared voltage-divider reference (SVDR) cell enables compact 2T1M storage, significantly improving memory density. Compared with [1] and [18], the proposed design achieves 73.96% and 64% improvements in area efficiency, respectively.

II. Proposed Bit-splitting MRAM-CIM Design

1) Architecture overview: **Fig. 2(a)** illustrates the fully parallel bit-splitting digital MRAM-CIM macro. The macro integrates 32 64×64-bit MRAM subarrays along with peripheral circuits including decoders, BL/SL drivers, input feature buffers, WL drivers, I/O circuits, and timing. Each subarray contains eight Partial MAC (PMAC#0–#7) units and one DM-based weight-wise adder tree, enabling parallel IN-8b × W-8b MAC operations within a single cycle. Each PMAC unit comprises 8×64 2T1M cells, eight DMLATs, and one DM-based input-wise adder tree. The macro supports both signed and unsigned INT-8b MAC modes.

2) CIM Operation: As shown in **Fig. 2(b)**, during weight access, both V_{IN} and WL_{REF} are asserted high to activate transistor N3, while RWL[0] and WWL[0] are enabled to turn on transistors N2 and N1. WBL is set to VSS. This establishes a conductive path between SVDR and the voltage-gated spin–orbit torque MTJ (VGSOT-MTJ[24]), as illustrated in **Fig. 3(a)**, forming a voltage-divider network (from N3, R_{REF} to N1, MTJ, N2) whose output is sensed by an inverter (INV) to generate the digital weight data. In this design, eight 2T1M cells share a single SVDR cell, enabling high-density storage. The accessed weights are then used to perform dot-product operations through a NOR gate. It is noteworthy that the accessed data represent the inverted form of the stored weights. The timing of this process is shown in **Fig. 3(b)**.

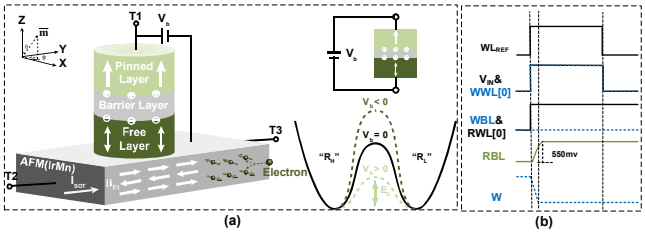


Fig. 3. (a) Schematic of the three-terminal VGSOT-MTJ. (b) Timing of dot-product operations.

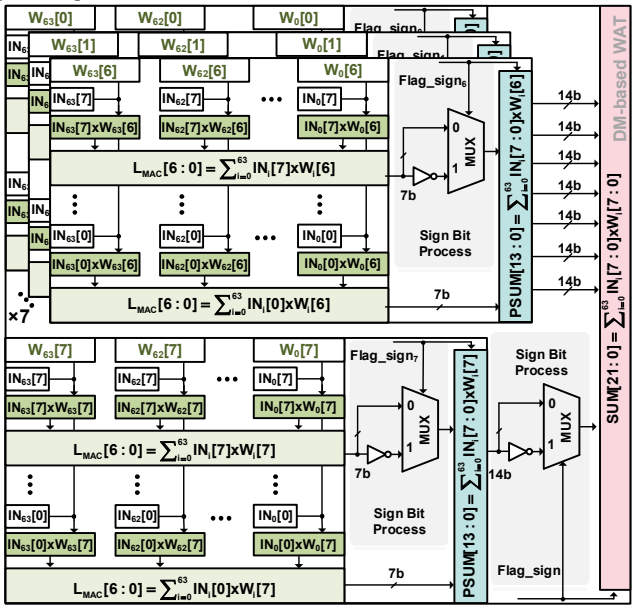


Fig. 4 Detailed bit-splitting MAC mapping diagram of IN-8b/W-8b/OUT-22b.

When the VGSOT-MTJ is in a high-resistance (R_H) or low-resistance (R_L) state, the resulting read bit-line (RBL) voltages (V_{RSL}) are V_H and V_L, respectively, with their difference

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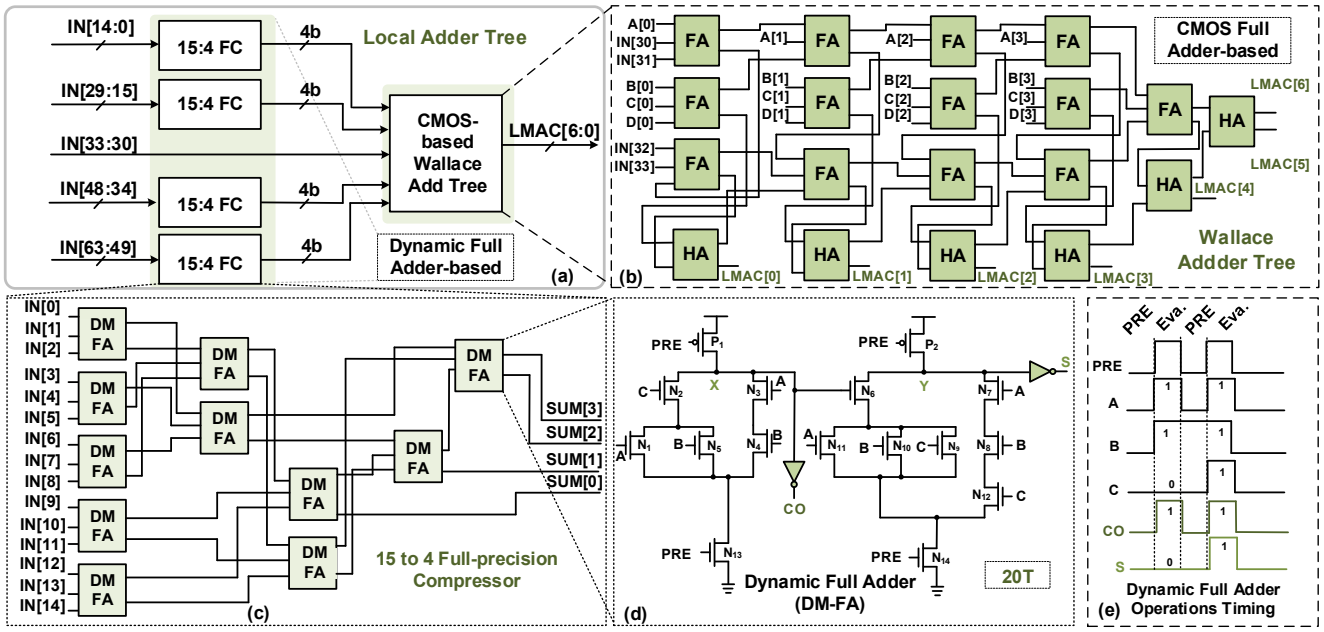


Fig. 5. (a) Schematic of the proposed dynamic local adder tree (DMLAT). (b) Schematic of the customized Wallace tree. (c) Schematic of the 15-to-4 full-precision compressor (FC). (d) Schematic of the dynamic full adder (DM-FA). (e) Timing diagram of the DM-FA.

defining the sensing margin. The corresponding INV-based sensing performance will be analyzed in **Section III**. Then, V_H or V_L is converted by the INV into digital logic '0' or '1'. After accessing weights from a subarray, the weight bits $W_{63-0}[0]$~$W_{63-0}[7]$ are multiplied bit-splitting with the parallel inputs $IN_{63-0}[7:0]$, as shown in **Fig. 4**.

Within each PMAC, all 64 groups of 2T1M cells are accessed simultaneously to read weights $W_{63-0}[0]$, and the resulting dot products $IN_{63-0}[i] \times W_{63-0}[0]$ are accumulated in the DMLAT to form 7-bit local MAC outputs $LMAC[6:0]$. The complete local accumulation is expressed as:

$$LMAC_{7-0}[6:0] = \sum_{i=0}^{7} \sum_{k=0}^{63} IN_k[i] \times W_k[0]. \quad (1)$$

Since the least significant bit (LSB) of weight is multiplied by the most significant bit (MSB) of input, $LMAC_7$ represents the sign-bit. To avoid complex sign-extension, a sign-detection is performed: if $LMAC_7[6:0]$ =0, Flag_sign is set to 0; otherwise (Flag_sign = 1), $LMAC_7$ is inverted and "+1" operation (2'complement conversion) using MUXs and INVs. The "+1" operation is completed within the DM-based weight-wise adder tree, enabling signed MAC without explicit sign-extension.

All 8-bit weights $W_{63-0}[7:0]$are read simultaneously, and the sign-processed $LMAC_7[6:0]$ and $LMAC_{6-0}[6:0]$ are accumulated by a DM-based input adder to produce 14-bit partial sums ($PSUM_{7-0}[13:0]$). $PSUM_7[13:0]$ undergo similar sign-bit processing as described previously. Finally, all $PSUM_{7-0}[13:0]$ results are accumulated to generate the final 22-bit output ($SUM[21:0]$).

$$SUM[21:0] = \sum_{j=0}^{7} \sum_{i=0}^{7} S_{i,j}\, 2^{i+j} \sum_{k=0}^{63} IN_k[i] \times W_k[j] \quad (2)$$

As depicted in **Eq. (2)**, the proposed bit-splitting MAC operation thus requires only simple sign-detection, avoiding complex sign-extension logic. Moreover, unsigned mode is enabled by setting Flag_sign = 0.

3) Dynamic adder tree design: As shown in **Fig. 5(a)**, the DMLAT consists of four 15-to-4 FCs and a customized Wallace tree (**Fig. 5(b)**), compressing unweighted inputs into the full-precision local MAC result $LMAC[6:0]$. Each 15-to-4 FC (**Fig. 5(c)**) cascades eleven DM-FAs. Because each stage in the 15-to-4 FC maintains full-bit logic utilization, the structure achieves much lower logic overhead compared to conventional CMOS-based adder trees [15]. This approach reduces approximately 121 HAs and 63 FAs with a 64-accumulation length, significantly minimizing logic depth and transistor count. The DM-FA (**Fig. 5(d)**) adopts a precharge–evaluate dynamic logic scheme that substantially improves speed and energy efficiency. During precharge (PRE = VSS), internal nodes are charged high; during evaluation (PRE = VDD), the NMOS network performs logic computation. For example, when PRE = 0, nodes X and Y remain high, causing both CO and S to output low levels. When PRE = VDD, the internal nodes discharge according to inputs A, B, and C. Specifically, when A = B = C = 1, nodes X and Y are pulled low during the evaluation phase, producing logic '1' outputs for S and CO. The corresponding timing is shown in **Fig. 5(e)**.

By exploiting dynamic node transition for sum and carry generation, redundant inversion and static current paths are eliminated, cutting transistor count by nearly 50%. Consequently, both power and area are greatly reduced.

Integrating DM-FAs into the 15-to-4 FC chain further enhances area and energy efficiency, making this structure particularly suitable for highly parallel and resource-constrained CIM systems.

III. Experimental Result Analysis

The proposed bit-splitting MRAM-CIM memory was simulated using a hybrid 28-nm CMOS and VGSOT-MTJ technology. The important parameters of the VGSOT-MTJ

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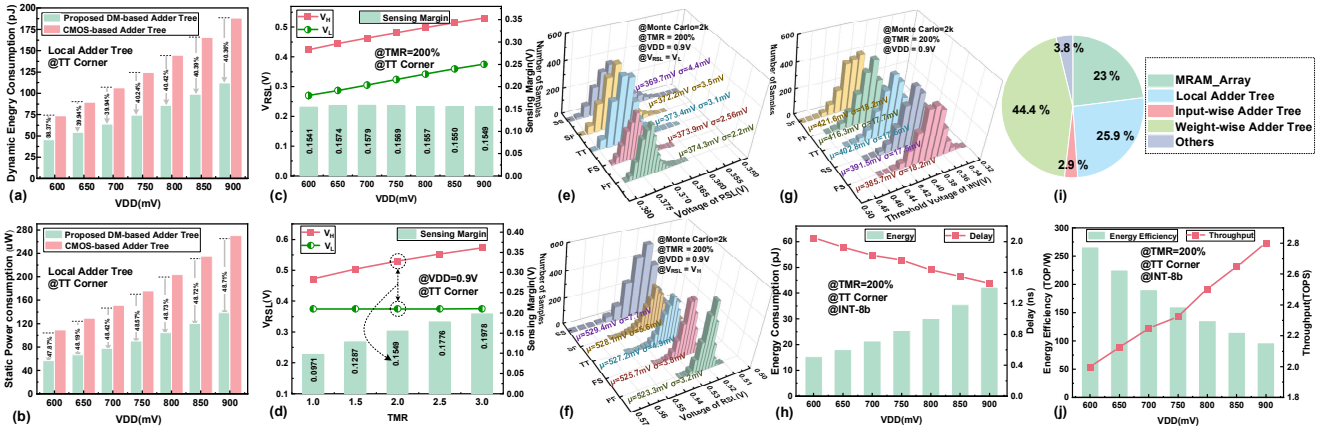


Fig. 6. (a) Dynamic energy and (b) static power comparison of the proposed DM-based local adder tree with its CMOS-based counterpart. Simulation results of V_{RSL} and sensing margin under different (c) VDDs and (d) TMRs. Monte Carlo simulation under different process corners: (e) V_L of RSL, (f) V_H of RSL, and (g) threshold voltage of INV. (h) Energy-delay simulation results of bit-splitting MAC. (i) Energy breakdown. (j) Energy efficiency and throughput simulation results of bit-splitting MAC.

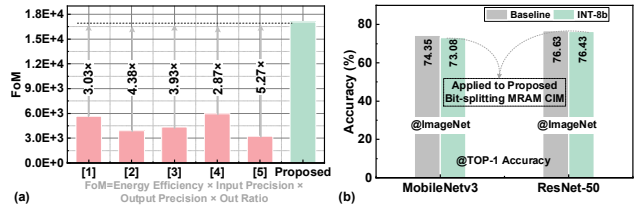


Fig. 7. (a) FoM comparison with state-of-the-art CIM macros. (b) Inference accuracy of the proposed MRAM-CIM macro.

devices are included in [17], [18]. In this study, 200% TMR was selected to achieve the proposed circuit design. The memory array consisted of 32 × 64 × 64 cells, corresponding to a total capacity of 128 kb.

The core adder tree of this design is constructed with a 15-to-4 FC, itself composed of DM-FAs. Thus, we compared the dynamic energy and static power of the DM-based and CMOS-based LATs. Results show a 40% reduction in dynamic power (**Fig. 6(a)**) and approximately 48% in static power (**Fig. 6(b)**). **Figs. 6(c)** and **(d)** present V_{RSL} and read margin under varying VDDs and TMRs. A TMR of 200% is shown to be sufficient for robust weight access. To further validate CIM operations, we conducted 2k Monte Carlo simulations for V_L, V_H, and INV threshold voltage (V_{TH}), as shown in **Figs. 6(e)–(g)**. V_{TH} remains consistently distributed near the midpoint between the mean values of V_L and V_H across process corners, confirming reliable access performance.

Fig. 6(h) illustrates energy consumption and delay of the CIM macro under different VDDs at TMR = 200%. As supply voltage increases, bit-splitting MAC operations exhibit higher energy consumption but reduced delay. The corresponding energy breakdown is provided in **Fig. 6(i)**.

Fig. 6(j) shows energy efficiency and throughput of the CIM macro. The proposed 128 kb CIM macro performs 32 groups of parallel MAC operations in one cycle, with multiplication and accumulation counted as 32 × 64 × 2 operations (OP). At VDD = 0.6 V and 0.9 V, the macro achieves energy efficiencies of 265.97 TOPS/W (calculated as 32×64×2 OP / 15.4 pJ) and 96.9 TOPS/W, respectively, with corresponding throughputs of 2.0 TOPS and 2.8 TOPS, at INT-8b. **Fig. 7(a)** presents a figure-of-merit (FoM) comparison with state-of-the-art CIM macros, demonstrating a 3.03 to 5.27× improvement. This work achieves 73.08% and 76.43% inference accuracy, respectively, when used on MobileNetv3 @ImageNet and ResNet-50 @ImageNet (**Fig. 7(b)**). **Table I** presents a comparison between the proposed work and prior studies.

TABLE I
COMPARISON OF THE PROPOSED STUDY WITH THE PREVIOUS WORKS

	This work	TCAS-I'25 [1]	ISSCC'24 [2]	ISSCC'25 [3]	TCAS-I'24 [4]	ISSCC'24 [5]
Technology	**28 nm CMOS/MTJ**	28 nm CMOS/MTJ	22nm CMOS	28nm CMOS	14nm CMOS/MTJ	28nm CMOS
Cell type	**2T1M**	4T2M	6T	6T	3T1M	6T
Array size	**128 kb**	288 kb	64 kb	64 kb	2 kb	192 kb
Supply voltage (V)	**0.6-0.9**	0.6-0.8	0.6-0.9	0.6-0.9	0.8	0.7-0.95
Comp. delay (ns)	**1.46 [7]-2.05 [6]**	3.1 [5]-6.2 [6]	4.1 [7]	12 [7]	36	4.8
W/IN precision	**8b/8b**	8b/8b	8b/8b	8b/8b	8b/8b	8b/8b
Out Ratio [3]	**1**	1	1	1	1	1
Access Strategy	**INV-based**	INV-based	Hybrid (ADC/NOR)	AND-based	RSA-TDC [12]	ADC
Computing Mode	**Parallel**	Serial (2b per cycle)	Serial (2b per cycle)	Serial (Bit-rotated)	Serial (1b per cycle)	Parallel
Signed Operation	**Bit-Splitting**	NA	2's Comp.	Sign-Extension	NA	Signed-XOR
Energy Efficiency (TOPS/W)	**96.9 [1,7]** **266.5 [1,6]**	58 [1,5] 88 [1,6]	20.7 [2,7] 60.8 [2,6]	21.04 [2,7] 67.8 [2,6]	92.9 [1,5]	22.78-50.53 [2,13]
Throughput (TOPS)	**2.0 [1,6]** **2.8 [1,7]**	1.15 [1,6] 1.49 [1,5]	0.062 [2,7]	0.04 [2,6] 0.17 [2,7]	45.5 [1,5] (INT-1b)	2.89-5.31 [2]
FoM [4]	**17056 [6]**	5632 [6]	3891.2 [6]	4339.2 [6]	5945.6 [5]	3233.92
Accuracy in Neural Network (%)	**73.08 [15]** **76.43 [16]**	98.92 [8] 93.33 [9]	69.25 [10] 80.03 [11]	68.48 [10] 78.49 [11]	89.56 [9]	75.26 [14]

[1]Simulation; [2]Chip measurement; [3]Actual out resolution / Ideal out resolution; [4]Energy efficiency×Input precision×Weight precision×Out ratio; [5]Performance @VDD=0.8V; [6]Performance @VDD=0.6V; [7]Performance @VDD=0.9V; [8]Performance @MNIST; [9]Performance @CIFAR-10; [10]ResNet-18 @ImageNet; [11]Vision transformer @ImageNet; [12]Resistance summation array-based TDC (RSA-TDC); [13]Performance @Average; [14]ResNet-18@CIFAR-100; [15]MobileNetv3@ImageNet (TOP-1 Accuracy); [16]ResNet-50 @ImageNet (TOP-1 Accuracy).

IV. CONCLUSION

This work presents a bit-splitting MRAM-CIM macro to address the limitations of logic overhead and sign-bit processing in DCIM. By implementing a dynamic adder tree and bit-splitting MAC scheme in a 28 nm process with VGSOT-MTJ devices, the 128-kb macro achieves 48.7% lower MAC energy and eliminates sign-extension. The design reaches 96.9 TOPS/W and 2.8 TOPS at INT-8b, demonstrating a 3.03–5.27× improvement in figure-of-merit over prior arts.

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